Random Variables and Expectation

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COMP2711 Discrete Mathematical Tools

1 Random variables

An outcome is one result of an experiment. An event is a set of outcomes. A random variable records a number from the outcome. Flip a fair coin three times independently. Instead of recording the full string, suppose we only want the number of heads.

MUST UNDERSTAND Random variable

A random variable is a function $X : S \to \mathbb{R}$ assigning a number to each outcome in the sample space S. The function is fixed; which outcome occurs is random. For the head count:

Outcomes	X
TTT	0
HTT, THT, TTH	1
HHT, HTH, THH	2
HHH	3

For example, $X(HTH) = 2$. The event $X = 2$ means the set $\{HHT, HTH, THH\}$, so $\Pr[X = 2] = 3/8$.

Examples.

Experiment	Number recorded
Throw two dice.	The sum of the results.
Draw five cards without replacement from 52 cards.	The number of aces.
Roll a fair die until the first 6.	The number of rolls, including the final 6.
Draw one card.	1 if it is a heart, and 0 otherwise.

If g is a real-valued function defined on the values of X, then $g(X)$ is also a random variable. Examples include X^2 and e^X. We can also combine variables on the same sample space: $X + Y$ and XY are random variables because each outcome determines their values.

2 Distributions

MUST UNDERSTAND Distribution

For a discrete random variable, its distribution lists each possible value and its probability. Write $p_X(x) = \Pr[X = x]$. Then

$$0 \le p_X(x) \le 1, \qquad \sum_x p_X(x) = 1.$$

Here and below the sums are over possible values of X. Several outcomes may give the same value; add their probabilities:

$$\Pr[X = x] = \sum_{\omega : X(\omega) = x} \Pr[\{\omega\}].$$

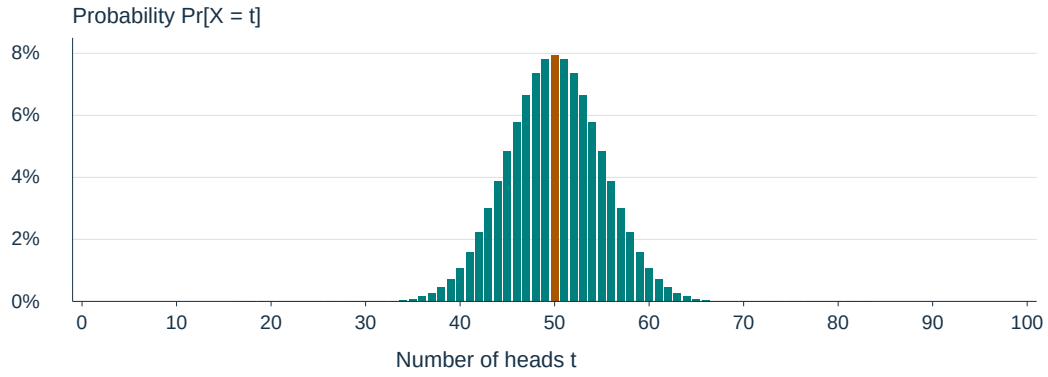
For the three-flip count,

x	0	1	2	3
$\Pr[X = x]$	1/8	3/8	3/8	1/8

The eight strings are equally likely; the four values are not. For a set A of possible values, add their probabilities:

$$\Pr[X \in A] = \sum_{x \in A} \Pr[X = x].$$

For example, the three-flip count satisfies $\Pr[X \ge 2] = 3/8 + 1/8 = 1/2$.



For 100 independent fair tosses, each bar gives the probability of one possible head count. Many outcomes produce counts near 50; very few produce counts near 0 or 100.

Independent random variables.

Discrete random variables X, Y are independent if and only if, for every pair of values x, y,

$$\Pr[X = x \text{ and } Y = y] = \Pr[X = x] \Pr[Y = y].$$

Learning the value of one does not change the distribution of the other. Equivalently, for every y with $\Pr[Y = y] > 0$ and every x, $\Pr[X = x \mid Y = y] = \Pr[X = x]$.

Roll two fair dice independently. Let X be the first result, Y the second, and $Z = X + Y$ their sum. For each $x, y \in \{1, \dots, 6\}$, $\Pr[X = x \text{ and } Y = y] = 1/36 = (1/6)(1/6)$, so X, Y are independent.

Pairs involving a value outside this range have zero probability on both sides. But X, Z are not independent:

$$\Pr[X = 1 \text{ and } Z = 12] = 0 \neq \Pr[X = 1]\Pr[Z = 12] = \frac{1}{6}\frac{1}{36}.$$

Indeed, learning $Z = 12$ forces $X = 6$, so its distribution changes. One failed pair disproves independence; one successful pair does not prove it.

2.1 Common discrete distributions

Distribution	Description
Uniform	n possible values, each with probability $1/n$.
Bernoulli	1 with probability p, and 0 otherwise.
Binomial	Number of successes in n independent trials with the same success probability p.
Geometric	Number of independent trials up to and including the first success, with fixed success probability $p > 0$.

Example: Guessing on a quiz.

A quiz has three questions, each with four options and exactly one correct answer. A student guesses independently and uniformly on each question. Let X count correct answers. Write C for correct and W for wrong; their probabilities are $1/4$ and $3/4$.

Exactly two correct answers can occur in three disjoint patterns:

$$\Pr[CCW] = \Pr[CWC] = \Pr[WCC] = \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{3}{64}.$$

Add these probabilities: $\Pr[X = 2] = 3(3/64) = 9/64$. In general, choose the k correct positions, then multiply by the probability of one pattern:

$$\Pr[X = k] = \binom{3}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{3-k}.$$

The full distribution is

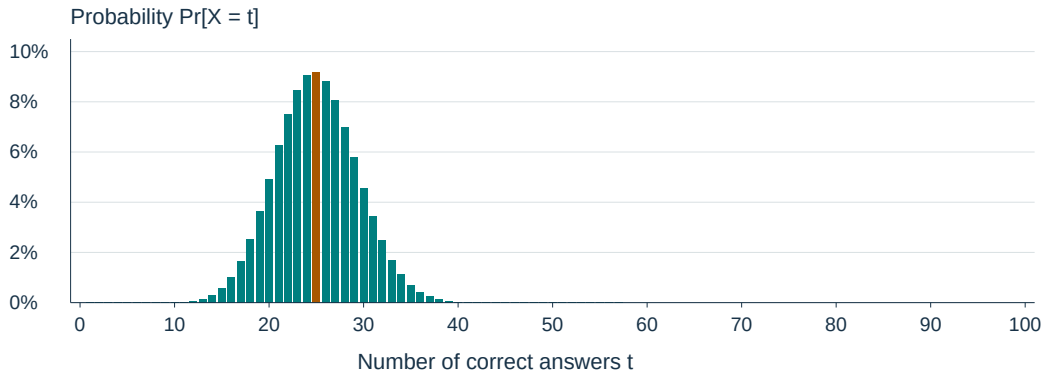
k	0	1	2	3
$\Pr[X = k]$	27/64	27/64	9/64	1/64.

Binomial distribution.

For n independent trials, each succeeding with probability p, let X count successes. One specified pattern with k successes has probability $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ choices of their positions, so

$$\Pr[X = k] = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

This is the **binomial distribution**, written $X \sim \mathrm{Bin}(n, p)$. The formula requires the same p and mutual independence. For $p = 0$ or 1, factors with exponent zero contribute one. Guessing on 100 four-option questions gives $X \sim \mathrm{Bin}(100, 1/4)$:



Drawing cards without replacement does not give independent trials. For example, exactly two aces in five cards has probability $\binom{4}{2}\binom{48}{3}/\binom{52}{5}$, not $\binom{5}{2}(1/13)^2(12/13)^3$. The latter applies with replacement and reshuffling.

3 Expectation

Example: A two-outcome game.

A game gives a net gain of \$20 with probability 1/4 and a net loss of \$5 with probability 3/4. What average gain should we expect over many independent plays?

In n plays, approximately $n/4$ are wins and $3n/4$ are losses. The average gain per play is therefore approximately

$$\frac{(n/4)20 + (3n/4)(-5)}{n} = \frac{1}{4}(20) + \frac{3}{4}(-5) = 1.25.$$

This suggests the rule: multiply each value by its probability, then add.

MUST UNDERSTAND Expectation is a probability-weighted average.

Denote the expectation of X by $\mathbb{E}[X]$. Its definition is

$$\mathbb{E}[X] = \sum_x x \Pr[X = x] = \sum_{\omega \in S} X(\omega) \Pr[\{\omega\}].$$

The first expression groups outcomes by value; the second sums over individual outcomes. The main examples here have finitely many values.

For the game, the calculation is:

Outcome	Gain	Probability	Gain × probability
Win	20	1/4	5
Lose	−5	3/4	−3.75

Thus $\mathbb{E}[X] = 5 - 3.75 = 1.25$ dollars per play. This is not a possible gain on a single play.

For many independent samples $X_1, \ldots, X_n$ from the same distribution, the average $(X_1 + \cdots + X_n)/n$ is likely to be close to $\mathbb{E}[X]$. This is the intuition behind the **law of large numbers**, made precise in the next lecture. It does not predict the next individual result.

Functions of a random variable.

For a function g,

$$\mathbb{E}[g(X)] = \sum_x g(x)\Pr[X = x],$$

provided the expectation exists. Transform each value, then take the weighted average. For the two-outcome game,

$$\mathbb{E}[X^2] = 20^2\frac{1}{4} + (-5)^2\frac{3}{4} = 118.75, \qquad (\mathbb{E}[X])^2 = 1.25^2 = 1.5625.$$

In general, $\mathbb{E}[g(X)]$ need not equal $g(\mathbb{E}[X])$. Similarly, $\mathbb{E}[XY]$ need not equal $\mathbb{E}[X]\mathbb{E}[Y]$; taking $Y = X$ in this game gives a counterexample.

3.1 Linearity of expectation

MUST UNDERSTAND Linearity does not require independence.

For random variables with finite expectations and constants a, b,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y], \qquad \mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

The sum of two fair dice has mean $7/2 + 7/2 = 7$.

To prove additivity in the discrete setting, sum over the same outcomes:

$$\mathbb{E}[X + Y] = \sum_\omega (X(\omega) + Y(\omega))\Pr[\{\omega\}] = \mathbb{E}[X] + \mathbb{E}[Y].$$

No product of probabilities is used, so no independence assumption is needed. Similarly,

$$\mathbb{E}[aX + b] = \sum_x (ax + b)\Pr[X = x] = a\mathbb{E}[X] + b\sum_x \Pr[X = x] = a\mathbb{E}[X] + b.$$

Expectation of a binomial distribution.

Let $X \sim \mathrm{Bin}(n, p)$. Directly from the definition,

$$\mathbb{E}[X] = \sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k}.$$

We can avoid evaluating this sum. Let I_j equal 1 if trial j is a success and 0 otherwise. Then $\mathbb{E}[I_j] = 1p + 0(1-p) = p$ and $X = I_1 + \cdots + I_n$. Linearity gives

$$\mathbb{E}[X] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = np.$$

Indicators.

For an event A, its indicator is

$$I_A = \begin{cases} 1, & A \text{ occurs,} \\ 0, & A \text{ does not occur.} \end{cases} \qquad \mathbb{E}[I_A] = 1\Pr[A] + 0\Pr[\overline{A}] = \Pr[A].$$

If X counts how many of $A_1, \ldots, A_n$ occur, then $X = \sum_i I_{A_i}$ and

$$\mathbb{E}[X] = \sum_i \Pr[A_i].$$

A random permutation.

Uniformly permute the numbers $1, \ldots, n$, where $n \geq 1$. How many numbers are expected to remain in their original positions? Let $I_i = 1$ when position i contains i. Each number is equally likely to occupy that position, so $\Pr[I_i = 1] = 1/n$. Thus

$$X = \sum_{i=1}^{n} I_i, \qquad \mathbb{E}[X] = \sum_{i=1}^{n} \frac{1}{n} = 1.$$

We do not need the full distribution of X or independence of the indicators. For example, with two numbers, either both stay or neither stays.

Optional: Waiting times and search

These applications use expectation and linearity; the waiting-time argument also uses an infinite geometric series.

The first success.

Repeat independent trials with fixed success probability $0 < p \leq 1$. Let G count trials up to and including the first success. Then

$$\Pr[G = k] = (1-p)^{k-1} p, \qquad k = 1, 2, \ldots.$$

This is the geometric distribution. The probabilities sum to one. Writing a positive integer as a sum of ones gives

$$\mathbb{E}[G] = \sum_{j=1}^{\infty} \Pr[G \geq j] = \sum_{j=1}^{\infty} (1-p)^{j-1} = \frac{1}{p}.$$

The first equality follows by adding the nonnegative terms in the other order. Waiting for the first 6 on a fair die has mean 6 rolls; it is not guaranteed to finish within 6 rolls.

Several successes.

Let W_r count trials needed for r successes, $r \geq 1$. Split the wait into r successive waits for one success. Each has mean $1/p$, so $\mathbb{E}[W_r] = r/p$. If a die roll is accepted when it is not 6, the mean number of rolls for ten accepted results is $10/(5/6) = 12$.

Average-case search.

A target occurs in a list of n distinct elements, with each position equally likely. Linear search takes $T = i$ comparisons when the target is at position i. Therefore

$$\mathbb{E}[T] = \frac{1}{n} \sum_{i=1}^{n} i = \frac{n+1}{2}.$$

The input distribution is part of an average-case claim.

Problem Set

1. Flip a fair coin twice independently. Let X count heads. Find its distribution and expectation. Find $\mathbb{E}[X(X+1)]$.

2. Five independent trials each succeed with probability 0.4. Find the probability of exactly two successes and the expected success count.

3. Draw five cards uniformly without replacement from 52 cards. What is the expected number of aces? Do not find the full distribution.

4. A uniformly random permutation of $1, \dots, 6$ is chosen. Find the expected number of positions containing their original numbers. Are the corresponding indicators independent? Explain.

5. Optional. Roll a fair die until the first 6. Find $\Pr[G > 4]$ and $\mathbb{E}[G]$. What is the mean number of rolls until the third 6?

6. Optional. A target is uniformly located among 99 distinct list entries. Find the expected number of comparisons in successful linear search.

Solutions

1. The probabilities for $0, 1, 2$ are $1/4, 1/2, 1/4$. Thus $\mathbb{E}[X] = 1$ and $\mathbb{E}[X(X+1)] = 0(1/4) + 2(1/2) + 6(1/4) = 5/2$.

2. $\Pr[X = 2] = \binom{5}{2}(0.4)^2(0.6)^3 = 0.3456$. Writing X as five success indicators gives $\mathbb{E}[X] = 5(0.4) = 2$.

3. Let I_i indicate that draw i is an ace. Each position is uniform among the 52 cards, so $\mathbb{E}[I_i] = 4/52$. By linearity the answer is $5(4/52) = 5/13$, despite dependence between draws.

4. Each position is correct with probability $1/6$, so the answer is 1. For distinct i, j, the probability both are correct is $(1/6)(1/5) = 1/30$, not $(1/6)^2$. Thus the indicators are not independent.

5. Four failures give $\Pr[G > 4] = (5/6)^4 = 625/1296$. The first-success mean is 6, and the third-success mean is $3/(1/6) = 18$.

6. $\mathbb{E}[T] = (99+1)/2 = 50$ comparisons.